

RENANCE

PETROLEUM TRAINING INSTITUTE, EFFURUN

SEC 113 — Past Questions & Materials

by Renance

Course: SEC 113

Institution: Petroleum Training Institute, Effurun

Papers in this pack: 1

Levels: ND1

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Paper 1 of 1 — SEC113 PRACTICAL - Hazard Identification in the Classroom and Workshop

Level ND1 • First Semester • Handout

COURSE CODE: SEC113 PRACTICAL

TITLE: Hazard Identification in the Classroom and Workshop

AIM

To provide students with practical skills to identify potential hazards in the classroom and workshop environment and assess their risk in order to prevent accidents.

OBJECTIVE

1. To recognize unsafe acts and unsafe conditions in the classroom.
2. To identify different types of hazards present in the classroom and workshop environment.
3. To understand how hazards can lead to accidents if not properly controlled.

MATERIALS

- Personal Protective Equipment (PPE)
- Sample incident report form
- Case study scenario

THEORY

Hazard Identification

Hazard identification is the process of recognizing and identifying potential sources of danger that may

cause injury, illness, or damage to property within a workplace, classroom, or workshop environment. It is

an important part of accident prevention because identifying hazards early helps to reduce risks and

improve safety.

A hazard is anything that has the potential to cause harm to a person, equipment, or the environment.

Types of Hazards

1. Physical Hazard

Physical hazards are environmental conditions that can cause injury to individuals without necessarily

involving direct contact with chemicals or electricity. Examples include wet floors, poor lighting, extreme

temperatures, noise, and obstructed pathways.

2. Ergonomic Hazard

Ergonomic hazards occur when the design of tools, equipment, or furniture does not suit the user, causing

discomfort, strain, or injury. Examples include broken chairs, poorly arranged desks, improper lifting

techniques, and repetitive movements.

3. Electrical Hazard

Electrical hazards arise from faulty electrical systems, exposed wires, damaged electrical equipment, or improper use of electrical appliances. These hazards can cause electric shock, burns, or fire outbreaks.

4. Chemical Hazard

Chemical hazards are substances that can cause harm when inhaled, ingested, or when they come in contact with the skin. Examples include improperly stored chemicals, cleaning agents, and laboratory chemicals.

DIAGRAM

Personal Protective Equipment (PPE)

- Safety Boots
- Goggles
- Gloves
- Helmet

(Students are expected to draw and label the PPE in their practical notebooks.)

PROCEDURE

The classroom and workshop environment were carefully inspected in order to identify possible hazards.

The inspection focused on the following conditions:

1. Loose electrical cords or cables
2. Wet floors
3. Obstructed pathways
4. Broken furniture
5. Improper storage of chemicals
6. Poor lighting and ventilation
7. Unsafe use of electrical equipment

During the inspection, hazards 1, 3, 4, 5, and 6 were found to be present in the laboratory.

Scenario (Reported Speech)

It was reported that during a laboratory practical session, a student attempted to move across the laboratory to collect some materials for an experiment. It was observed that some loose electrical cables

were lying across the floor, which created an electrical hazard in the laboratory.

As the student walked through the laboratory, it was reported that some equipment and boxes had been

left on the pathway, which obstructed the walkway. While trying to avoid the obstruction, the student

accidentally tripped over the loose electrical cable.

It was also reported that as the student tried to support himself, he leaned on a broken chair, which collapsed due to its poor condition. At the same time, a nearby shelf containing improperly stored

laboratory chemicals was hit, causing some chemical containers to fall.

It was further reported that the situation became more difficult because the lighting and ventilation in the

laboratory were poor, making it difficult for other students to quickly identify the hazards and assist immediately.

This incident demonstrated how the presence of multiple hazards in a laboratory environment could combine to create a serious accident if proper safety measures are not followed.

IDENTIFICATION OF HAZARDS

Hazard Observed Type of Hazard

Loose electrical cords or cables Electrical Hazard

Obstructed pathway Physical Hazard

Broken furniture Ergonomic Hazard

Improper storage of chemicals Chemical Hazard

Poor lighting and ventilation Physical Hazard

UNDERLYING CAUSES

- Poor maintenance of electrical equipment
- Lack of proper housekeeping in the laboratory
- Failure to repair damaged furniture
- Improper storage and handling of chemicals
- Inadequate lighting and ventilation
- Lack of safety awareness among students and staff

PREVENTIVE MEASURES

- Electrical cables should be properly arranged and covered to prevent tripping or electrical shock.
- Pathways should always be kept clear of obstacles.
- Broken furniture should be repaired or replaced immediately.
- Chemicals should be properly stored in labeled containers and appropriate storage cabinets.
- Adequate lighting and ventilation should be provided in the laboratory.
- Safety training should be provided to students and staff.

OBSERVATION

During the inspection of the laboratory, several unsafe conditions were identified. Loose electrical cables

were seen on the floor, pathways were obstructed with materials, and some furniture was damaged.

Chemicals were not properly arranged or stored, and the lighting and ventilation were inadequate. These

conditions could easily lead to accidents such as electric shock, trips, falls, or chemical exposure.

PRECAUTION

- Personal Protective Equipment (PPE) should always be worn in the laboratory.
- Electrical equipment should be handled carefully.
- Chemicals should be handled and stored properly.
- Students should follow laboratory safety rules at all times.

RECOMMENDATION

- Regular safety inspections should be conducted in classrooms and laboratories.
- Faulty electrical systems should be repaired immediately.
- Damaged furniture should be replaced.
- Chemicals should be stored according to laboratory safety standards.
- Proper lighting and ventilation systems should be installed.

CONCLUSION

Hazard identification is an essential part of accident prevention in the classroom and workshop environment. Identifying hazards such as loose electrical cables, obstructed pathways, broken furniture, improper chemical storage, and poor lighting can help reduce the risk of accidents. Maintaining a safe and organized environment helps protect students, staff, and equipment.

EXERCISE

1. Define hazard identification.
2. List four types of hazards.
3. Mention three hazards that can be found in a classroom environment.
4. State two preventive measures for electrical hazards.
5. Why is PPE important in the laboratory or workshop?